

GLASGOW, MT, USA

WMO: 727680

Lat: 48.2067N

Lon: 106.6256W

Elev: 694

StdP: 93.26

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94008

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-25.9	-32.0	0.2	-28.5	-28.7	0.3	-25.3	14.4	-5.1	12.7	-8.7	3.2	310	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	34.8	18.3	32.9	17.8	30.8	17.2	20.5	29.7	19.4	28.8	18.5	27.8	4.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.7	23.9	16.3	12.6	22.5	15.2	11.7	21.8	62.2	29.4	58.2	28.7	54.8	27.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.3	10.0	-32.9	38.3	3.6	1.4	-35.5	39.3	-37.6	40.1	-39.6	40.9	-42.3	41.9		
(5)			-32.3	22.6	3.0	1.5	-34.5	23.7	-36.2	24.6	-37.8	25.5	-40.0	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-9.9	-8.1	-0.9	6.7	12.5	17.6	22.2	21.4	15.3	7.1	-1.2	-8.2		
	DBStd	12.78	8.51	8.00	7.82	5.18	4.63	3.80	3.36	3.79	4.89	5.19	6.99	7.81		
	HDD10.0	2644	617	507	342	122	27	1	0	0	11	116	338	563		
	HDD18.3	4700	875	741	597	349	188	57	6	16	115	348	587	822		
	CDD10.0	1287	0	0	2	22	104	229	378	352	171	27	1	0		
	CDD18.3	302	0	0	0	0	7	36	126	109	24	1	0	0		
	CDH23.3	3999	0	0	0	19	142	469	1596	1381	367	24	0	0		
	CDH26.7	1737	0	0	0	3	33	153	749	651	144	4	0	0		
(14) Wind	WSAvg	4.7	4.2	4.2	5.0	5.4	5.3	4.9	4.6	4.7	4.6	4.7	4.3	4.1		
(15) Precipitation	PrecAvg	287	10	7	11	20	42	61	47	33	23	16	9	9		
	PrecMax	451	32	25	32	50	95	136	151	146	105	78	39	26		
	PrecMin	171	0	1	1	2	6	2	0	0	1	0	0	0		
	PrecStd	71	7	5	6	13	24	34	37	31	23	15	7	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	11.5	20.2	26.1	30.1	33.8	37.5	37.2	33.8	26.7	18.0	9.4		
		MCWB	4.4	5.3	9.8	12.4	16.0	18.6	18.8	18.3	17.2	13.8	9.4	4.1		
	2%	DB	5.4	7.0	15.6	22.1	27.0	30.5	35.2	34.7	30.8	21.7	13.7	5.9		
		MCWB	2.2	2.9	7.6	11.1	14.8	17.9	18.7	18.0	15.5	11.6	7.1	2.1		
	5%	DB	3.3	4.2	12.4	19.0	24.5	28.1	33.2	32.7	27.7	18.4	10.3	3.7		
		MCWB	1.0	1.4	5.9	9.4	13.6	16.9	18.5	17.4	15.1	10.2	5.1	1.0		
	10%	DB	1.3	2.0	9.3	16.3	21.8	26.0	31.0	30.6	24.6	15.5	7.7	1.8		
		MCWB	-0.4	0.2	4.3	8.1	12.2	16.0	17.9	16.9	14.0	8.8	3.7	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	5.8	10.1	13.2	17.3	21.5	22.6	21.0	19.1	14.7	10.1	4.4		
		MCDB	8.1	10.8	18.8	23.8	26.1	30.1	32.1	31.6	28.6	23.9	16.8	8.6		
	2%	WB	2.4	3.1	8.1	11.5	15.6	19.2	21.0	19.6	17.1	12.4	7.5	2.5		
		MCDB	5.0	6.4	15.2	20.8	24.4	27.5	29.8	29.7	26.6	20.4	13.5	5.5		
	5%	WB	1.1	1.5	6.3	10.1	14.4	17.9	20.0	18.7	15.8	10.6	5.4	1.1		
		MCDB	3.2	4.0	11.8	17.9	22.5	25.9	29.1	28.9	25.3	17.3	9.5	3.4		
	10%	WB	-0.3	0.2	4.5	8.6	13.1	16.8	19.0	17.8	14.6	9.3	3.9	-0.1		
		MCDB	1.4	2.0	8.9	15.3	20.3	24.3	28.0	27.9	23.9	14.8	7.5	1.8		
(35) Mean Daily Temperature Range	MDBR		9.7	10.1	11.2	13.2	13.3	12.9	15.0	15.2	14.2	12.4	10.8	9.6		
	5% DB	MCDBR	11.0	12.2	16.1	18.3	17.4	16.5	18.1	18.1	18.9	17.2	14.8	11.6		
		MCWBR	8.4	8.6	9.2	9.3	7.9	7.0	6.2	6.3	7.7	8.9	9.2	8.6		
	5% WB	MCDBR	10.9	11.6	15.7	16.9	15.6	14.7	15.7	15.8	16.7	16.1	14.2	11.1		
		MCWBR	8.5	8.4	9.3	8.9	7.4	6.9	6.4	5.9	7.2	8.6	9.0	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.256	0.272	0.318	0.341	0.350	0.360	0.368	0.322	0.283	0.258	0.241		
	taud		2.498	2.540	2.577	2.439	2.419	2.445	2.430	2.380	2.478	2.585	2.558	2.520		
	Ebn at Noon		857	931	963	935	921	911	895	872	888	884	842	828		
	Edh at Noon		59	72	82	104	111	109	109	110	90	68	56	51		
(44) All-Sky Solar Radiation	RadAvg		1.39	2.36	3.62	4.95	5.96	6.56	7.01	5.83	4.22	2.69	1.60	1.08		
	RadStd		0.11	0.17	0.25	0.24	0.45	0.40	0.34	0.38	0.34	0.23	0.17	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page